

Constructing Determined Sets from below

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1 Panorama of Determined Sets

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Recall that $\mathbb{R} = \omega^\omega$.

One of the main ways to compare the complexity of sets of reals is the *modified continuous reducibility*, a.k.a. the *modified Wadge-reducibility*:

$A \leq_w^* B$ iff for some continuous $f : \mathbb{R} \rightarrow \mathbb{R}$, $A = f^{-1}[B]$ or $A = f^{-1}[B^c]$.

We will be interested in the preorder $(\mathcal{P}(\mathbb{R}), \leq_w^*)$.

Definition

$\Gamma \subseteq \mathcal{P}(\mathbb{R})$ is a *pointclass* iff for all $A \in \Gamma$ and all $B \leq_w^* A$, $B \in \Gamma$.

If Γ is a pointclass, then $\Gamma / =_w^* = \{[A]_w^* : A \in \Gamma\}$.

Also, $\Gamma / =_w^*$ is downwards-closed in $(\mathcal{P}(\mathbb{R}), \leq_w^*) / =_w^*$.

Notation

$\mathcal{D}$ denotes the pointclass of all sets $A \subseteq \mathbb{R}$ such that $L(\mathbb{R}, A) \models \text{AD}^+$.

Remark

Consistently, $\mathcal{D} = \emptyset$.

Fact (Martin, Steel, Woodin)

If there is a limit of Woodin cardinals with a measurable above it, then there is an order-preserving injection $\omega_1 \hookrightarrow \mathcal{D} / \equiv_w^$.*

Theorem (Martin, Wadge)

$(\mathcal{D}, \leq_w^*) / =_w^*$ is a tree.

Question

Is $\mathcal{D} / =_w^*$ linear?

Theorem (Woodin)

It is consistent, modulo large cardinals, that $\mathcal{D} / =_w^*$ is **not** linear.

Definition

A pointclass Γ is *closed* iff $\mathcal{P}(\mathbb{R}) \cap L(\mathbb{R}, \Gamma) = \Gamma$.

Fact

There is a bijective correspondence between closed pointclasses and inner models M such that $\mathbb{R} \subseteq M$ and $M \models "V = L(\mathcal{P}(\mathbb{R}))"$. The correspondence is given by $\Gamma \leftrightarrow L(\Gamma)$.

Definition

The pointclass $\mathcal{D}_{\mathbb{R}}$ is defined as the union of all closed pointclasses Γ such that $L(\Gamma) \models \text{AD}^+ + \text{AD}_{\mathbb{R}}$.

Fact

$\mathcal{D}_{\mathbb{R}} / =_{\mathbb{W}}^*$ is a sub-tree of $\mathcal{D} / =_{\mathbb{W}}^*$.

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Question

How can you “construct” sets in $\mathcal{D}$?

Example: every projective set of reals is the solution set of a first-order formula over the structure $(H_{\omega_1}, \in, \mathbb{R})$.

Reminder: $\{\text{projective sets}\} = \mathcal{P}(\mathbb{R}) \cap L_1(\mathbb{R})$.

Example: every set of reals in $L_2(\mathbb{R})$ is the solution set of a first-order formula over the structure $(L_1(\mathbb{R}), \in)$.

Example: every set of reals in $L(\mathbb{R})$ can be obtained by iterating the definable-powerset operation, i.e. $L_1(-)$.

$\mathcal{D}_0 := \mathcal{P}(\mathbb{R}) \cap L(\mathbb{R})$ is a pointclass.

$\mathcal{D}_0 / \equiv_w^*$ has a supremum in $\mathcal{D} / \equiv_w^*$.

How to construct some/any representative of this equivalence-class?

Definition

Let X be a set. We denote by μ_X the club-filter on $[X]^\omega$.

Definition

Let Γ be a pointclass. We define the pointclass $\mathcal{O}_1(\Gamma)$ to be

$$\mathcal{P}(\mathbb{R}) \cap L(\mathbb{R}, \Gamma^\omega)[\mu_\Gamma].$$

Fact (Woodin)

$$\mathcal{A}_0 \subset \mathcal{O}_1(\mathcal{A}_0) \subseteq \mathcal{D}.$$

Theorem

Suppose Γ is a pointclass which is a **bounded** subset of $\mathcal{D}_{\mathbb{R}}$. Then:

- a. (Solovay, essentially) $\Gamma \subseteq \mathcal{O}_1(\Gamma) \subseteq \mathcal{D}_{\mathbb{R}}$;
- b. (Ikegami-Trang) If $\mathcal{O}_1(\Gamma) = \Gamma$, then Γ is a closed pointclass and $L(\Gamma) \models \text{AD}_{\mathbb{R}} + \text{DC}$.

Corollary (Originally, theorem by Woodin)

Suppose there exists an inner model M such that $\mathbb{R} \subseteq M$ and $M \models \text{AD}_{\mathbb{R}} + \text{DC}$. Then there exists an ordinal α such that $L(\mathcal{O}_1^\alpha(\mathcal{D}_0))$ is the $\subseteq$ -minimum such model.

Fact

*Suppose M is an inner model such that $\mathbb{R} \subseteq M$ and $M \models \text{AD}_{\mathbb{R}} + \text{“}\Theta \text{ is regular”}$. Then there exists a pointclass Γ such that Γ is a **bounded** subset of $\mathcal{P}(\mathbb{R}) \cap M$ and $\mathcal{O}_1(\Gamma) = \Gamma$. In particular, there is **no** ordinal α such that $\mathcal{O}_1^\alpha(\mathcal{A}_0) = \mathcal{P}(\mathbb{R}) \cap M$.*

Definition

We say that a pointclass Γ is *of Chang type* iff

$$L(\Gamma^\omega, \text{Ord}^\omega)[\mu_\Gamma, (\mu_\alpha : \alpha < \text{Ord})] \cap \mathcal{P}(\mathbb{R}) = \Gamma.$$

Fact

If Γ is **not** of Chang-type, then there exists an ordinal α such that $L(\Gamma^\omega, \alpha^\omega)[\mu_\Gamma, \mu_\alpha]$ contains a set of reals which is **not** in Γ .

Definition

Let Γ be a pointclass which is **not** of Chang-type. Then we denote by κ_Γ the least α as above and by $\mathcal{O}(\Gamma)$ the powerset of reals in $L(\Gamma^\omega, \alpha^\omega)[\mu_\Gamma, \mu_\alpha]$.

Theorem (K., Sargsyan)

Suppose M is a $\subseteq$ -minimal inner model such that $\mathbb{R} \subseteq M$ and $M \models \text{AD}_{\mathbb{R}} + \text{“}\Theta \text{ is regular”}$. Let Γ be a pointclass which is a **bounded** subset of $\mathcal{P}(\mathbb{R}) \cap M$.

Then:

- a. Γ is **not** of Chang-type;
- b. $\Gamma \subsetneq \mathcal{O}(\Gamma) \subseteq M$.

Corollary

Suppose there is an inner model such M such that $\mathbb{R} \subseteq M$ and $M \models \text{AD}_{\mathbb{R}} + \text{“}\Theta \text{ is regular”}$. Then there is an ordinal α such that $L(\mathcal{O}^\alpha(\mathcal{D}_0))$ is the $\subseteq$ -least such M .

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Definition

The Chang-plus model, denoted by CM^+ , is the inner model

$$L(\text{Ord}^\omega)[(\mu_{\alpha^\omega} : \alpha \in \text{Ord})].$$

Theorem (Woodin; Ikegami-Trang)

Assume there are cofinally-many Woodin cardinals that are limits of Woodin cardinals. Then $\text{CM}^+ \models \text{AD}_{\mathbb{R}} + \text{DC} + “\omega_1 \text{ is supercompact}”$.

Theorem (K., Sargsyan)

Assume there are cofinally-many Woodin cardinals that are limits of Woodin cardinals. There exists a closed pointclass Γ which is a bounded subset of $\mathcal{P}(\mathbb{R}) \cap \text{CM}^+$ such that $L(\Gamma) \models \text{AD}_{\mathbb{R}} + “\Theta \text{ is regular}”$.

Woodin actually showed that there is a *universally Baire* sharp for CM^+ .

This means that every set of reals in CM^+ is universally Baire.

It also means that there an inner model M such that $\mathcal{P}(\mathbb{R}) \cap \text{CM}^+ \subseteq M$, $M \models \text{AD}_{\mathbb{R}}$, and furthermore, every set of reals in M is universally Baire.

It follows from the work of Sargsyan and Trang that one of possibilities A and B occurs, where:

A: there is a strategy for a hod mouse with a measurable limit of Woodin cardinals in CM^+ ;

B: there is a strategy for some hod mouse which is in M , but not in CM^+ .

If A occurs, then, in particular, there is a hod mouse in CM^+ that satisfies ZFC+“there is an inaccessible limit of Woodin cardinals”.

If A does not occur, then, in particular, B occurs, and there is a *minimal* pair of a hod mouse with its strategy which is not in CM^+ . Call it (P, Σ) .

It follows from Woodin's work on CM^+ that $\mathcal{P}(\mathbb{R}) \cap \text{CM}^+$ is of Chang-type.

Now, the proof of K.-Sargsyan Theorem shows that P has a measurable limit of Woodin cardinals.

This means that some *smaller* hod mouse (Q, T) satisfies $\text{ZFC} +$ “there is an inaccessible limit of Woodin cardinals”.

But every smaller hod mouse belongs to CM^+ !

Therefore, in every case, there is a hod mouse in CM^+ that satisfies $ZFC +$ “there is an inaccessible limit of Woodin cardinals”.

Now, Gappo and Sargsyan showed that the derived model of such a mouse at the inaccessible limit of Woodin cardinals is a model of $AD_{\mathbb{R}} + “\Theta$ regular”.

After doing the genericity iteration, we obtain the desired pointclass from (the image of) the derived model.

QED

Thank you for your attention!